

When are the strong-extension engines exact?

A formal analysis of the credal-network boundary case, and the exact characterization of exactness for Credal VE and Interval BP

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Abstract

The credal-network engines that bound the *strong extension* $\mathcal{D}_{\text{SE}}$ — Credal VE (exact over $\mathcal{D}_{\text{SE}}$) and Interval BP (outer over $\mathcal{D}_{\text{SE}}$; exact on trees) — are known to be exact with respect to the LCN’s true distribution set $\mathcal{D}_{\text{LCN}}$ on a singly-connected LCN *with no cross-family sentence*, where “cross-family” is judged by whether a sentence’s atom scope fits inside a single family scope. This note shows that atom-scope cross-family membership is *not* the condition that actually governs exactness. We exhibit a shipped singly-connected tree, `benchmarks/tree_small/tree_n5.1.lcn`, on which every sentence is atom-scope family-local yet Credal VE and Interval BP are strictly loose ($P(x_2) = [0, 0.8012]$ against the exact $[0.010, 0.4395]$). We identify the true governing property — a sentence is *family-realizable* iff the quantity it bounds is a function of one family’s *conditional* $P(\text{ch}_v \mid \text{pa}_v)$ alone, rather than a function that also depends on the parents’ marginal $P(\text{pa}_v)$ — and prove the sharp characterization: with the “**linear**” local sets, $\mathcal{D}_{\text{SE}} = \mathcal{D}_{\text{LCN}}$ (equivalently Credal VE, and Interval BP on a tree, is exact for every atom) *if and only if* the primal graph is singly connected and every sentence is family-realizable. The familiar “no cross-family sentence” condition is exactly the atom-scope-visible half of family-realizability; the invisible half has two forms, both atom-scope family-local: (a) a Type-1 formula mentioning a non-root child (a bare marginal $P(x)$ or a child–parent joint $P(x \wedge \text{parent})$), and (b) a Type-2 sentence whose condition ψ is a *formula* satisfied by two or more parent configurations — the *collider* case $P(x_2 \mid x_1 \oplus x_3)$, which binds through the co-parents’ joint (their LMC independence). Crucially, this looseness is a property of the strong extension, not the solver: we confirm that neither the certified global SCIP solver nor the LMC-tightening scheme D1 recovers the bound, because the coupling cannot be represented as a product of per-parent-config conditional credal sets; the exact route is scheme D2 (merge) or the junction-tree engine **CredalJT** (D5). We give the full `tree_n5.1` derivation, minimal witnesses (including the solver-invariant collider), a remark that a non-realizable sentence can even mask an inconsistency, a complete case catalogue over (sentence shape) $\times$ (node role), and the engine-specific consequences and fixes.

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1 Setup

We recap only what we need; see `docs/properties.tex` and `docs/cve_exactness.tex` for the full framework.

An LCN over binary atoms $\mathbf{V}$ has a set of *sentences* (Type-1 $\ell \leq P(\varphi) \leq u$ or Type-2 $\ell \leq P(\varphi \mid \psi) \leq u$) and the Local Markov Condition (LMC) of its primal graph; together they define the *LCN distribution set* $\mathcal{D}_{\text{LCN}} \subseteq \Delta(2^{\mathbf{V}})$ of joints satisfying every sentence and every LMC equality. The chain-graph factorization splits the joint into *families* $P(\text{ch}_v \mid \text{pa}_v)$, one per node v of the simplified structure graph; the family’s *scope* is $\text{ch}_v \cup \text{pa}_v$. Root families have $\text{pa}_v = \emptyset$, so their factor is the plain marginal $P(\text{ch}_v)$.

For each family the `cn/` pipeline (`lcn/inference/marginal/cn/local_credal_sets.py, method="linear"`) computes a *local credal set* K_v : the set of conditional tables $P(\text{ch}_v \mid \text{pa}_v)$ attainable by *some* joint that satisfies the sentences whose atom scope lies inside the family scope. Operationally it is a linear-fractional program $P(\text{ch}_v, \text{pa}_v)/P(\text{pa}_v)$ solved by the Charnes–Cooper transformation, in which *the parents’ distribution* $P(\text{pa}_v)$ *is a free variable* — the program returns the range of the conditional, not of any marginal. The engines then bound the *strong extension*

$$\mathcal{D}_{\text{SE}} = \left\{ \prod_v P_v : P_v \in K_v \text{ chosen independently per parent-config} \right\}, \quad (1)$$

and, because a chain-graph product satisfies every LMC assertion,

$$\mathcal{D}_{\text{LCN}} \subseteq \mathcal{D}_{\text{SE}}. \quad (2)$$

Credal VE returns *exactly* the strong-extension interval $[q_{\text{SE}}, \bar{q}_{\text{SE}}]$ (`docs/cve_exactness.tex`); Interval BP is an outer bound over $\mathcal{D}_{\text{SE}}$ that is exact on a tree factor graph (`docs/ibp_exactness.tex`). Both consume the same enumerated vertices of the K_v , so both are exact with respect to $\mathcal{D}_{\text{LCN}}$ for a queried atom *iff* $\mathcal{D}_{\text{SE}} = \mathcal{D}_{\text{LCN}}$ there. This note characterizes exactly when that holds.

2 Family-realizable sentences

The key is which sentences a single family’s local set can *enforce*. A family’s degrees of freedom are the conditional probabilities $P(\text{ch}_v=c \mid \text{pa}_v=p)$; the parents’ marginal $P(\text{pa}_v)$ is not one of them (it is fixed elsewhere in the network, and free during the local solve).

Definition 1 (Family-realizable sentence). *A sentence s is family-realizable in family v if the probability it bounds is a function of the conditional table $P(\text{ch}_v \mid \text{pa}_v)$ alone — i.e. its value is unchanged by varying the parents’ distribution $P(\text{pa}_v)$. Concretely this holds iff s is one of:*

- a Type-1 sentence $P(\varphi)$ on a root family ($\text{pa}_v = \emptyset$, so $P(\varphi)$ is a function of $P(\text{ch}_v)$); or
- a Type-2 sentence $P(\varphi \mid \psi)$ with $\text{atoms}(\varphi) \subseteq \text{ch}_v$ where the condition ψ pins a single, full parent-configuration — i.e. exactly one assignment of pa_v satisfies ψ . Then $P(\varphi \mid \psi)$ is a single row of the conditional table, independent of $P(\text{pa}_v)$.

A sentence is non-realizable otherwise. Note the Type-2 clause is subtler than “ $\text{atoms}(\psi) \subseteq \text{pa}_v$ ”: if ψ is a formula satisfied by two or more parent configurations (e.g. $x_1 \oplus x_3$), then $P(\varphi \mid \psi) = \sum_{c \models \psi} P(\varphi, c) / \sum_{c \models \psi} P(c)$ is a $P(\text{pa}_v)$ -weighted mixture of conditional rows — a function of the parents’ joint distribution, hence non-realizable even though $\text{atoms}(\psi) \subseteq \text{pa}_v$ (Remark 1, Example 3). An LCN is fully realizable if every sentence is family-realizable in some family.

Remark 1 (Atom scope is not enough). *Family-realizability is strictly stronger than the atom-scope “fits in a family” test. A Type-1 sentence $P(\varphi)$ whose formula φ mentions a non-root child x has atoms that may well sit inside a family scope, yet it bounds a marginal or joint of p , not a conditional. Since for a non-root node $P(\varphi) = \sum_{\text{pa}} P(\varphi \mid \text{pa}) P(\text{pa})$ depends on the parents’ marginal, it is not a function of any one family’s conditional and is therefore non-realizable. Two canonical offenders, both atom-scope family-local:*

- *a **bare marginal on a non-root child**, $P(x)$ with $\text{pa}_x \neq \emptyset$ (atoms $\{x\} \subseteq$ every family containing x);*
- *a **child-parent joint**, $P(x \wedge y)$ with $y = \text{pa}_x$ (atoms $\{x, y\} =$ the family scope of $x \mid y$), which equals $P(x \mid y) P(y)$ and again couples the conditional to $P(y)$;*
- *a **multi-configuration- ψ Type-2 sentence at a collider**, $P(\varphi \mid \psi)$ where ψ is a formula satisfied by ≥ 2 parent configurations (e.g. $P(x_2 \mid x_1 \oplus x_3)$ with co-parents x_1, x_3). Its value mixes conditional rows weighted by $P(\text{pa}_v)$; equivalently it binds through the co-parents’ joint, which for a collider is governed by the co-parent LMC assertion $x_1 \perp\!\!\!\perp x_3$ — a joint constraint the per-parent-config credal set cannot carry.*

This is not a solver, tightening-scheme, or enumeration artifact. We verified that neither the global SCIP backend nor scheme D1 (*“linear-tight”*, which does add the co-parent LMC bilinear equality to the family solve) recovers the bound: the offending family’s per-parent-config extreme points are genuinely the full box (each individual row is unconstrained; only a $P(\text{pa}_v)$ -weighted mixture is), so any representation of the family as a product of independent per-config conditional credal sets loses the coupling. The bound is recoverable only by keeping the coupled atoms in one joint block — scheme D2 (merge the family with its parents) or the exact junction-tree engine *CredalJT* (D5), which is precisely why D5 stays exact here while *Credal VE* / *Interval BP* cannot.

3 The exactness characterization

Theorem 1 (Exactness of the strong-extension engines). *Let $\mathcal{L}$ be a chain-graph LCN whose local credal sets are built by the “linear” method. Then $\mathcal{D}_{\text{SE}} = \mathcal{D}_{\text{LCN}}$ — equivalently, *Credal VE* (and *Interval BP* on the tree factor graph) is exact with respect to $\mathcal{D}_{\text{LCN}}$ for every atom — if and only if*

- (i) *the primal graph is singly connected, and*
- (ii) *every sentence is family-realizable (Definition 1).*

Proof. Sufficiency. Assume (i) and (ii). Fix a family v . Because every sentence constraining atoms inside $\text{ch}_v \cup \text{pa}_v$ is family-realizable, it bounds a functional of the conditional $P(\text{ch}_v \mid \text{pa}_v)$ only; hence K_v , the range of that conditional over joints meeting these sentences, is precisely the projection of $\mathcal{D}_{\text{LCN}}$ onto the family’s conditional (no Gap A on the imposed conditional: nothing that constrains the conditional was omitted, and nothing imposed constrains anything but the conditional). Now take any product $\prod_v P_v$ with $P_v \in K_v$. Being a chain-graph product it satisfies every LMC assertion by construction; and every sentence, being a function of a single family’s conditional P_v with $P_v \in K_v$, is satisfied. Therefore $\prod_v P_v \in \mathcal{D}_{\text{LCN}}$, giving $\mathcal{D}_{\text{SE}} \subseteq \mathcal{D}_{\text{LCN}}$; with (2), $\mathcal{D}_{\text{SE}} = \mathcal{D}_{\text{LCN}}$. (Single connectedness guarantees the primal graph moralizes to a chain graph whose families are exactly these factors, so the product form is the general element of $\mathcal{D}_{\text{LCN}}$.)

Necessity. Suppose (ii) fails: some sentence s is non-realizable in its family v , i.e. the quantity g it bounds is a genuine function $g(P(\text{ch}_v \mid \text{pa}_v), P(\text{pa}_v))$ that depends on $P(\text{pa}_v)$. The local set K_v

is built with $P(\text{pa}_v)$ free, so it retains every conditional table $P(\text{ch}_v \mid \text{pa}_v)$ that is consistent with s under *some* choice of $P(\text{pa}_v)$ — including tables that violate s under the parents’ *actual* network marginal. In the free product (1) the vertex chosen for v and the marginal $P(\text{pa}_v)$ induced by the other families are selected independently, so $\mathcal{D}_{\text{SE}}$ contains a product joint on which g falls outside $[\ell_s, u_s]$; that joint is not in $\mathcal{D}_{\text{LCN}}$. Hence $\mathcal{D}_{\text{SE}} \supsetneq \mathcal{D}_{\text{LCN}}$, and the affected marginal’s strong-extension range strictly contains its $\mathcal{D}_{\text{LCN}}$ range, so the engines are strictly loose there. The failure of (i) (a loop in the primal graph) makes $\mathcal{D}_{\text{SE}} \not\supseteq \mathcal{D}_{\text{LCN}}$ for the standard reason (the free product cannot reproduce the correlation carried by a second path); see `docs/properties.tex`. Constructive witnesses for the (ii) direction are Examples 1 (a non-root marginal) and 3 (a multi-configuration collider condition, whose looseness is invariant to solver and to scheme D1). $\square$

Corollary 1 (Refinement of “no cross-family sentence”). *The condition “singly connected and no cross-family sentence” used in the companion notes is exactly the atom-scope-visible special case of Theorem 1: an atom-scope cross-family sentence is always non-realizable, but so is a Type-1 formula mentioning a non-root child even though its atoms are family-local. Family-realizability is the sharp condition; “no cross-family sentence” is sufficient for the sentences it can see and silent about the rest.*

Remark 2 (A non-realizable sentence can mask an inconsistency). *Because a non-realizable sentence is dropped from every family’s enforceable set, $\mathcal{D}_{\text{SE}}$ can be non-empty even when $\mathcal{D}_{\text{LCN}} = \emptyset$. Adding M1: $P(x_1) \leq 0.25$ to the chain of Example 1 makes $\mathcal{D}_{\text{LCN}}$ empty (the transitions force $P(x_1) \geq 0.4$), which the exact engine **CredalJT** flags as degenerate ($[0, 1]$ everywhere), yet **Credal VE** returns the non-empty strong-extension bound $P(x_0) = [0.3, 0.5]$. The strong-extension engines therefore cannot be used to certify (in)consistency in the presence of non-realizable sentences.*

4 Worked example: tree_n5_1.lcn

The instance (a saved 5-atom tree) is

$$\begin{array}{ll} \text{s0: } 0.159 \leq P(\neg x_3) \leq 0.759 & \text{s4: } 0.01 \leq P(x_0 \mid \neg x_2) \leq 0.323 \\ \text{s1: } 0.351 \leq P(x_4 \mid \neg x_3) \leq 0.951 & \text{s5: } 0.01 \leq P(x_2) \leq 0.4395 \\ \text{s2: } 0.639 \leq P(\neg x_2 \mid x_3) \leq 1.0 & \text{s6: } 0.01 \leq P(x_3) \leq 0.533 \\ \text{s3: } 0.317 \leq P(\neg x_1 \mid x_4) \leq 0.917 & \end{array}$$

with primal tree $x_3 \rightarrow x_2 \rightarrow x_0$ and $x_3 \rightarrow x_4 \rightarrow x_1$ (Figure 1, left). Interval BP runs on the *bipartite factor graph* that the **cn/** pipeline builds from the chain-graph families — one factor node per family, connected to the variable nodes in its scope (Figure 1, right); the offending factor $f_2 = P(x_2 \mid x_3)$, which carries the non-realizable marginal **s5**, is highlighted.

The five families and the realizability of their attached sentences:

family	role	attached sentences	realizable?
$P(x_3)$	root	s0, s6 (marginals on root x_3)	yes
$P(x_4 \mid x_3)$	non-root	s1 (Type-2 $x_4 \mid x_3$)	yes
$P(x_1 \mid x_4)$	non-root	s3 (Type-2 $x_1 \mid x_4$)	yes
$P(x_0 \mid x_2)$	non-root	s4 (Type-2 $x_0 \mid x_2$)	yes
$P(x_2 \mid x_3)$	non-root	s2 (Type-2), s5: $P(x_2)$	no (s5)

Exactly one sentence is non-realizable: **s5**, a marginal on the *non-root* child x_2 . By Theorem 1 the instance fails condition (ii), so the strong-extension engines are loose, and the looseness is localized to the family carrying the non-realizable sentence and its descendants.

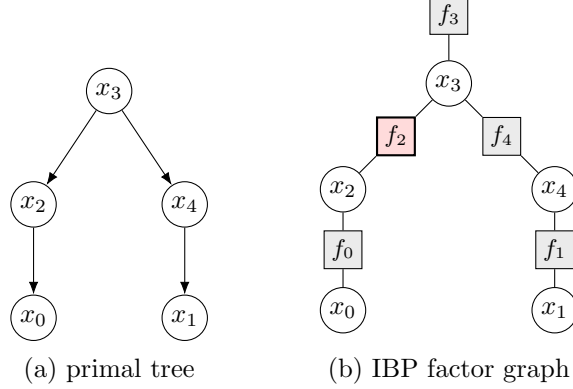


Figure 1: `tree_n5_1.1cn`. (a) The primal directed tree. (b) The bipartite factor graph Interval BP propagates over: circles are variable (atom) nodes, squares are factor nodes, one per chain-graph family — $f_3=P(x_3)$ (root prior), $f_2=P(x_2 | x_3)$, $f_4=P(x_4 | x_3)$, $f_0=P(x_0 | x_2)$, $f_1=P(x_1 | x_4)$ — each joined to the variables in its scope. The factor graph is a tree, so a correct interval BP would be exact over $\mathcal{D}_{SE}$; the looseness comes entirely from factor f_2 (shaded), whose non-realizable marginal **s5**: $P(x_2) \leq 0.4395$ cannot be enforced locally, so the composed $P(x_2)$ escapes to $[0, 0.8012]$ and inflates x_0 downstream through f_0 .

The witness for x_2 . The local set for $P(x_2 | x_3)$ enforces **s2** (a conditional) but not **s5** (the marginal). Composing its extreme points with those of $P(x_3)$ — the free product of (1) — the reachable marginal $P(x_2=1) = \sum_{x_3} P(x_2=1 | x_3) P(x_3)$ ranges over $[0, 0.8012]$, grossly exceeding **s5**’s cap 0.4395. The exact engines respect **s5**: **CredalJT** and **ARIEL** both return $P(x_2) = [0.010, 0.4395]$, hitting the sentence bound. **Credal VE** and **Interval BP** return exactly the strong-extension interval $[0.000, 0.8012]$.

Downstream propagation to x_0 . x_0 ’s family $P(x_0 | x_2)$ carries only the Type-2 **s4** on the $x_2=0$ branch, leaving $P(x_0 | x_2=1)$ vacuous. Since $P(x_0) = P(x_0 | x_2=0)P(x_2=0) + P(x_0 | x_2=1)P(x_2=1)$, the inflated $P(x_2)$ range combines with the free $x_2=1$ row and lifts x_0 from the exact $[0.0056, 0.6206]$ to the strong-extension $[0.002, 0.8654]$. The other three atoms are exact:

atom	Credal VE / Interval BP ($\mathcal{D}_{SE}$)	CredalJT / ARIEL (exact)	verdict
x_0	$[0.0020, 0.8654]$	$[0.0056, 0.6206]$	loose
x_1	$[0.0135, 0.9480]$	$[0.0135, 0.9480]$	exact
x_2	$[0.0000, 0.8012]$	$[0.0100, 0.4395]$	loose
x_3	$[0.2408, 0.5328]$	$[0.2408, 0.5328]$	exact
x_4	$[0.1639, 0.9771]$	$[0.1639, 0.9771]$	exact

x_1, x_3, x_4 are exact because their families carry only realizable sentences (root marginals on x_3 ; Type-2 conditionals for $x_4 | x_3$ and $x_1 | x_4$), matching Theorem 1: the looseness is confined to the non-realizable family $\{x_2 | x_3\}$ and its descendant x_0 .

5 Minimal witnesses

Example 1 (Marginal on a non-root child). *The two-atom chain $x_0 \rightarrow x_1$:*

$$P0: 0.3 \leq P(x_0) \leq 0.5; \quad T0: 0.2 \leq P(x_1 | x_0) \leq 0.4; \quad T1: 0.6 \leq P(x_1 | \neg x_0) \leq 0.8; \quad M1: 0.45 \leq P(x_1) \leq 0.55.$$

M1 is a marginal on the non-root child x_1 : non-realizable. Credal VE returns $P(x_1) = [0.40, 0.68]$ (ignoring M1, which its family $P(x_1 | x_0)$ cannot impose), while the exact bound is $[0.45, 0.55]$; x_0 stays exact $[0.30, 0.50]$. Removing M1 restores exactness on both atoms. This is the smallest instance exhibiting the boundary.

Example 2 (Child–parent joint). *With x_2 a root and x_0 its child:*

$$P2: 0.3 \leq P(x_2) \leq 0.6; \quad T0: 0.2 \leq P(x_0 | x_2) \leq 0.5; \quad T1: 0.1 \leq P(x_0 | \neg x_2) \leq 0.3; \quad J: 0.05 \leq P(x_0 \wedge x_2) \leq 0.15.$$

J has atoms $\{x_0, x_2\}$ — the entire scope of family $x_0 | x_2$ — so it is atom-scope family-local, yet it bounds $P(x_0 \wedge x_2) = P(x_0 | x_2) P(x_2)$, coupling the conditional to $P(x_2)$: non-realizable. Credal VE returns $P(x_0) = [0.13, 0.42]$ against the exact $[0.13, 0.36]$; x_2 stays exact $[0.30, 0.60]$. This shows the offender need not be a bare marginal, and confirms the atom-scope test cannot detect the case (Remark 1).

Example 3 (Collider with a multi-configuration condition). *A single collider $x_0 \rightarrow x_2 \leftarrow x_1$ with two roots and one Type-2 sentence whose condition spans both parents:*

$$P0: 0.3 \leq P(x_0) \leq 0.5; \quad P1: 0.4 \leq P(x_1) \leq 0.6; \quad C: 0.2 \leq P(x_2 | x_0 \oplus x_1) \leq 0.5.$$

*Here $\text{atoms}(\psi) = \{x_0, x_1\} \subseteq \text{pa}_{x_2}$, so C passes the naive Type-2 test, but $\psi = x_0 \oplus x_1$ is satisfied by two parent configurations, so C is non-realizable (Definition 1). The exact bound (**ExactInference** with the global SCIP solver, both endpoints certified at gap 0; **CredalJT** agrees) is $P(x_2) = [0.092, 0.77]$, $P(x_0) = [0.30, 0.50]$, $P(x_1) = [0.40, 0.60]$. Credal VE and Interval BP return $P(x_2) = [0, 1]$ — fully vacuous — because the x_2 family’s per-parent-config extreme points are each the full box $[0, 1]$ (no single row is constrained; only the $\{x_0 \neq x_1\}$ mixture is), so no product of per-config conditional sets can reproduce the bound. Neither the global solver nor scheme D1 (which adds the co-parent LMC equality $x_0 \perp\!\!\!\perp x_1$ to the family solve) changes this: on the D1 credal network, **CredalJT** is exact ($[0.092, 0.77]$) yet Credal VE is still $[0, 1]$, isolating the loss to the per-parent-config representation, not the solve. The bound returns only by fusing $\{x_0, x_1, x_2\}$ into one joint block (D2 merge) or via the joint NLP of **CredalJT** (D5). This is the witness underlying the loose atom in `benchmarks/polytree_small_fr/polytree_fr_n5_6.lcn`.*

6 The exactness catalogue

Table 1 enumerates the cases; “realizable?” is by Definition 1 and the verdict follows from Theorem 1 (assuming, for the sentence-shape rows, an otherwise singly-connected, fully-realizable instance so the row’s sentence is the only variable).

For every “loose (Gap B)” row the looseness is a property of the strong extension itself, *not* of the solver or of scheme D1: we verified (Example 3) that the certified global SCIP solver and “**linear-tight**” both leave the affected marginal loose, because the coupling is unrepresentable as a product of per-parent-config conditional credal sets. The exact route for these rows is scheme D2 (merge the coupled atoms into one joint block) or the junction-tree engine **CredalJT** (D5) / **ExactInference** (global).

Corollary 2 (Sufficient clean regimes). *A Markov chain, rooted tree, or polytree whose sentences are all root marginals and conditional-on-parent Type-2 sentences is fully realizable and singly*

Table 1: When the strong-extension engines (Credal VE; Interval BP on a tree) are exact w.r.t. $\mathcal{D}_{\text{LCN}}$.

Sentence shape / structural feature	realizable?	$\mathcal{D}_{\text{SE}}=\mathcal{D}_{\text{LCN}}?$	engine verdict
Type-2 $P(\varphi(\text{ch}) \mid \psi)$, ψ pins a <i>single</i> parent-config	yes	yes	exact
Type-1 $P(\varphi)$ on a <i>root</i> family	yes	yes	exact
Type-2 $P(\varphi \mid \psi)$, ψ a <i>multi-config</i> formula (collider)	no	no	loose (Gap B)
Type-1 $P(\varphi)$ mentioning a <i>non-root</i> child	no	no	loose (Gap B)
Type-1 child-parent joint $P(x \wedge \text{pa}_x)$	no	no	loose (Gap B)
atom-scope cross-family sentence (e.g. <code>d4.biting</code>)	no	no	loose (Gap B)
loopy primal graph (any sentences)	—	no	loose

connected; by Theorem 1 Credal VE and Interval BP are exact on it for every atom. This is the regime the benchmark generator now targets: `lcn/benchmarks/generator.py` emits its extra marginal sentences only on root atoms, so generated tree/polytree instances are fully realizable. Hand-authored instances (like `tree_n5_1.lcn`) may contain a non-realizable sentence and require the exact engine.

Engine-specific consequences and fixes.

- **Credal VE** is exact over $\mathcal{D}_{\text{SE}}$ unconditionally, so on a non-realizable instance it reports precisely the (inflated) strong-extension interval — a valid outer bound on $\mathcal{D}_{\text{LCN}}$, never unsound, but not tight.
- **Interval BP** inherits the same $\mathcal{D}_{\text{SE}}$ interval on a tree (it consumes the same vertices), hence the *identical* loose numbers; on a loopy factor graph it adds its own outer slack, and it performs no upward evidence update (`docs/ibp_exactness.tex`).
- **Restoring exactness** on a non-realizable instance requires moving the coupling into a single local solve: scheme D2 (`merge_budget`) merges the offending family with its parent so the marginal becomes enforceable in the merged joint, or the exact junction-tree engine **CredalJT** (scheme D5) solves the joint directly. Neither D1 (`linear-tight`) nor the atom-scope D4 helps, since the coupling is genuinely across families and invisible to the atom-scope test.

The companion notes give the per-engine detail: `docs/cve_exactness.tex`, `docs/ibp_exactness.tex`, and the consolidated `docs/properties.tex` (the master and topology tables), whose “clean singly-connected, cross-family-sentence-free” rows are, by Corollary 1, the fully-realizable instances of Theorem 1.